

# **ECON 100B: Macroeconomics**

## **Class 5: A Model of Production 2 (Jones Chapter 4)**

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**February 3, 2026**

# Agenda

- Announcements
- Current events
- Finish up Chapter 4: A Model of Production

# Announcements

- Plagiarism+ Quiz is due on bCourses this Friday 2/6
- Problem Set 1 is due on Gradescope on Friday 2/13
- Looking ahead:
  - Problem Set 2 will be due on Gradescope on Friday 2/27
  - In-class Midterm the next Thursday 3/5
  - Take-home Midterm Essay released 3/8, due 3/20
  - Spring Break 3/23-3/27

# Current events

- Plenty. Duh
- **Kevin Warsh** is the President's pick to be the next chair of the Federal Reserve Board of Governors
  - Harvard Macroeconomist [Greg Mankiw thinks this is good](#)
  - CUNY Macroeconomist & Nobelist [Paul Krugman is less sanguine](#)

# February is Black History Month

- [Lisa D. Cook](#) (b.1964) is a Ph.D. graduate (1997) of the University of California, Berkeley
- She is the first black woman to sit on the Federal Reserve Board of Governors, appointed in 2022
- Prior to joining the Fed, she was professor of economics at Michigan State
- Her Ph.D. work focused on banking in Russia
- Berkeley is lucky to have so many superb economics alumni. Also check out [Trevon Logan](#) (2004) and [Adriana Kugler](#) (1997)



# Outline of Macroeconomics

- Chapter 2 shows how we measure income,  $Y = C + I + G + NX$
- **Chapters 3–6 discuss long-run growth**
- Chapters 7–8 talk about inflation and unemployment
- Chapters 9–14 discuss short-run fluctuations, fiscal policy, monetary policy, the Phillips Curve, the Fed
- Chapters 16 & 17 focus on Consumption and Investment
- Chapter 18 looks at Government
- Chapters 19 & 20 address international trade and finance, exchange rates

# Outline of Macroeconomics

- Chapter 2 shows how we measure income,  $Y = C + I + G + NX$
- **Chapters 3–6 discuss long-run growth**
  - Chapter 3 introduced the growth rate rules

$$g[X \times Y] = g[X] + g[Y]$$

$$g[X/Y] = g[X] - g[Y]$$

$$g[X^a \times Y^b] = a \cdot g[X] + b \cdot g[Y]$$

- Chapter 4 introduces Cobb-Douglas production

$$Y = A K^{1/3} L^{2/3}$$

with constant returns to scale  
and declining marginal productivity of K or L

- Chapter 5 develops the Solow Model
- Chapter 6 develops the Romer Model



# Learning objectives in Chapter 4

- There are enormous differences in GDP per person,  $y$ , across countries
- The Cobb-Douglas production function  $Y = A K^\alpha L^{1-\alpha}$  can help us model GDP. GDP per worker is  $y = Y/L$
- This production function is realistic
  - Constant returns to scale:  
If you duplicate a factory and its workers, you duplicate output
  - Diminishing returns to any one factor:  
Increasing capital without increasing workers leaves idle capital
- What is technology,  $A$ ? If we assume it's the same across countries, we find a striking result: The model explains some of reality but far from all. Later, we will need to understand  $A$

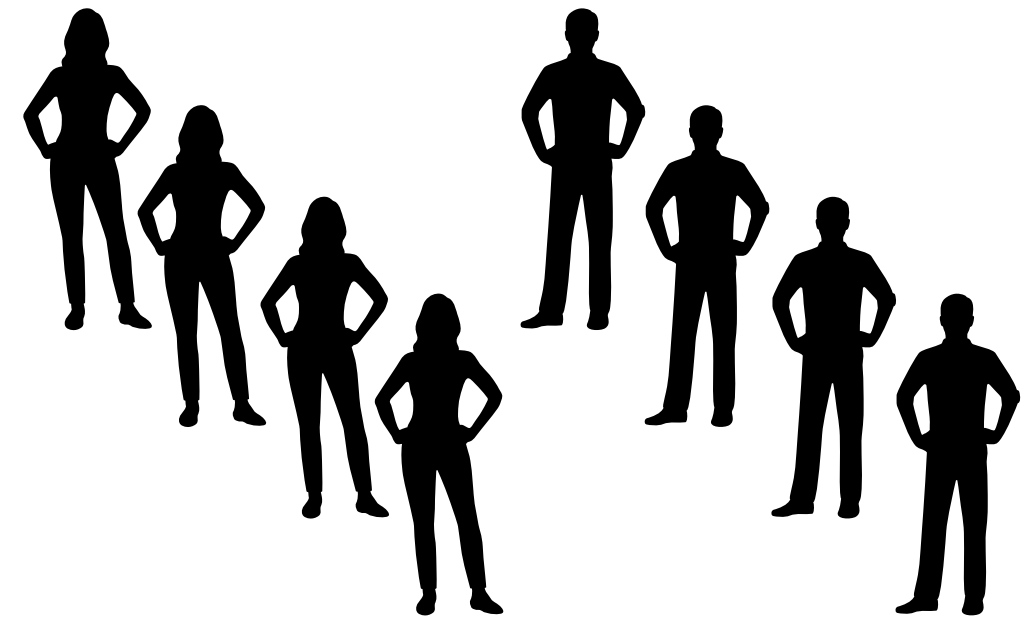
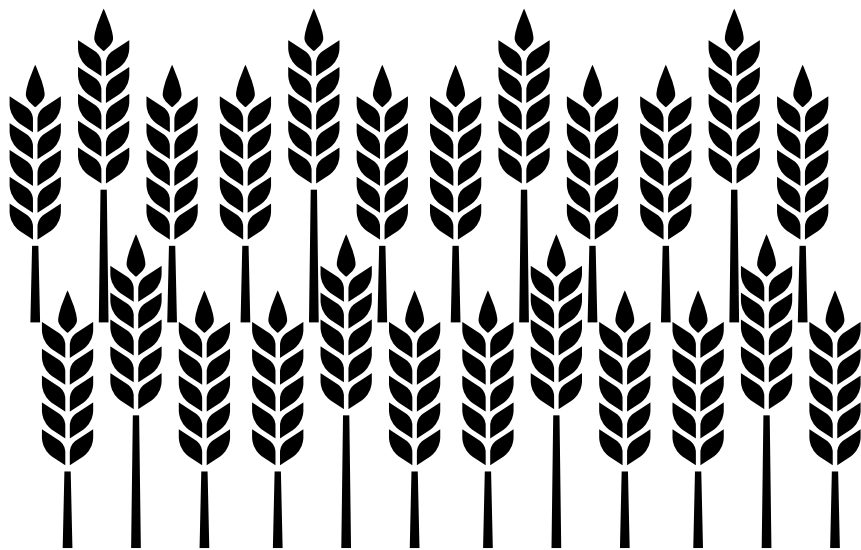


# Calibrated example of what we're talking about

- “Calibrated” = Real-world numbers that fit reality well

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- “Calibrated” = Real-world numbers that fit reality well
- Consider a **small farm** in 2026 with  $L = 8$  workers



# Capital goods make workers productive

- Farming requires land and machinery



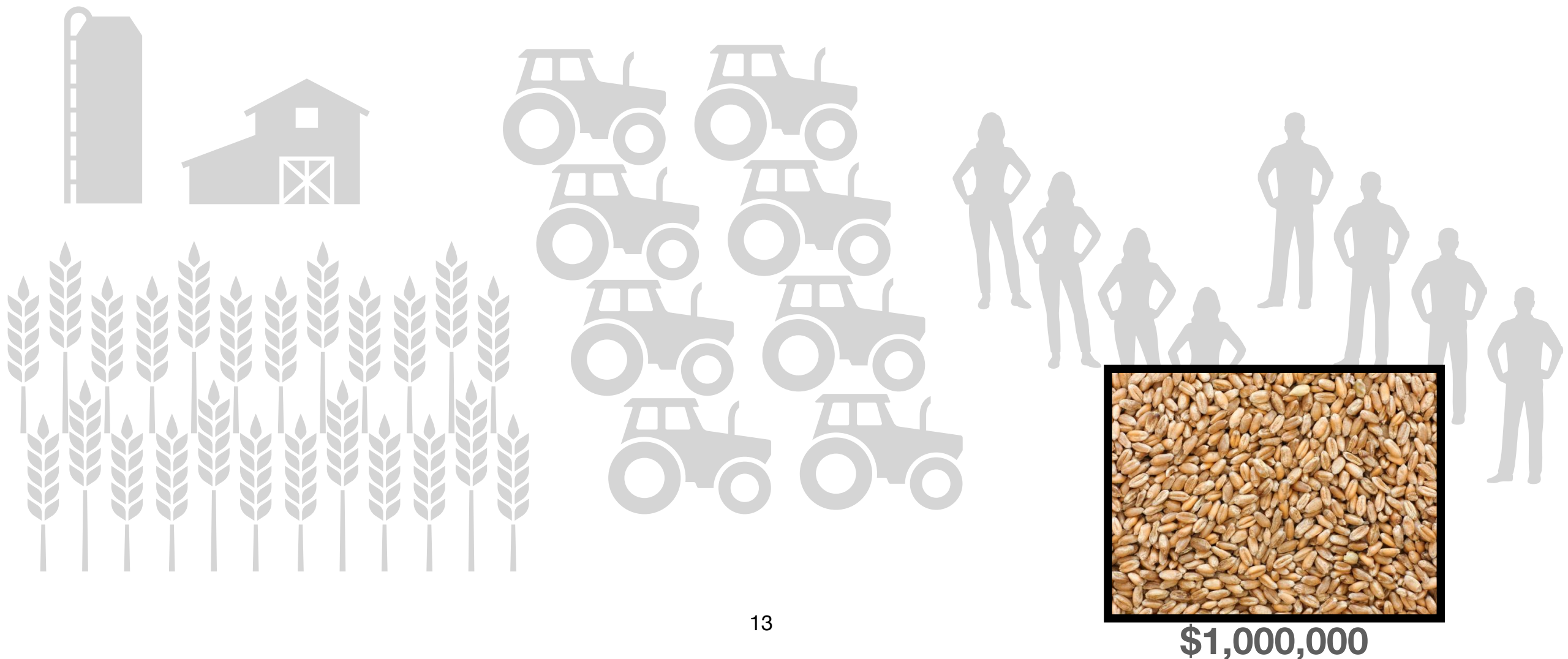
# In the U.S., the value of capital per worker is pretty high

- Farming requires land and machinery
- The total market value of all land and machinery that you see here might be  $K = \$4,000,000$
- Capital per worker is  $K/L = \$4,000,000 \div 8 = \$500,000$



# It turns out that often, $K/Y \approx 4$

- See [Piketty and Zucman \(QJE, 2014\)](#). The ratio has fluctuated
- With  $K = \$4,000,000$  we typically find  $Y = \$1,000,000$
- In words, with \$4 million in capital and 8 workers, this farm produces \$1 million worth of product each year



# Summarizing

$$K = \$4,000,000$$

$$L = 8$$

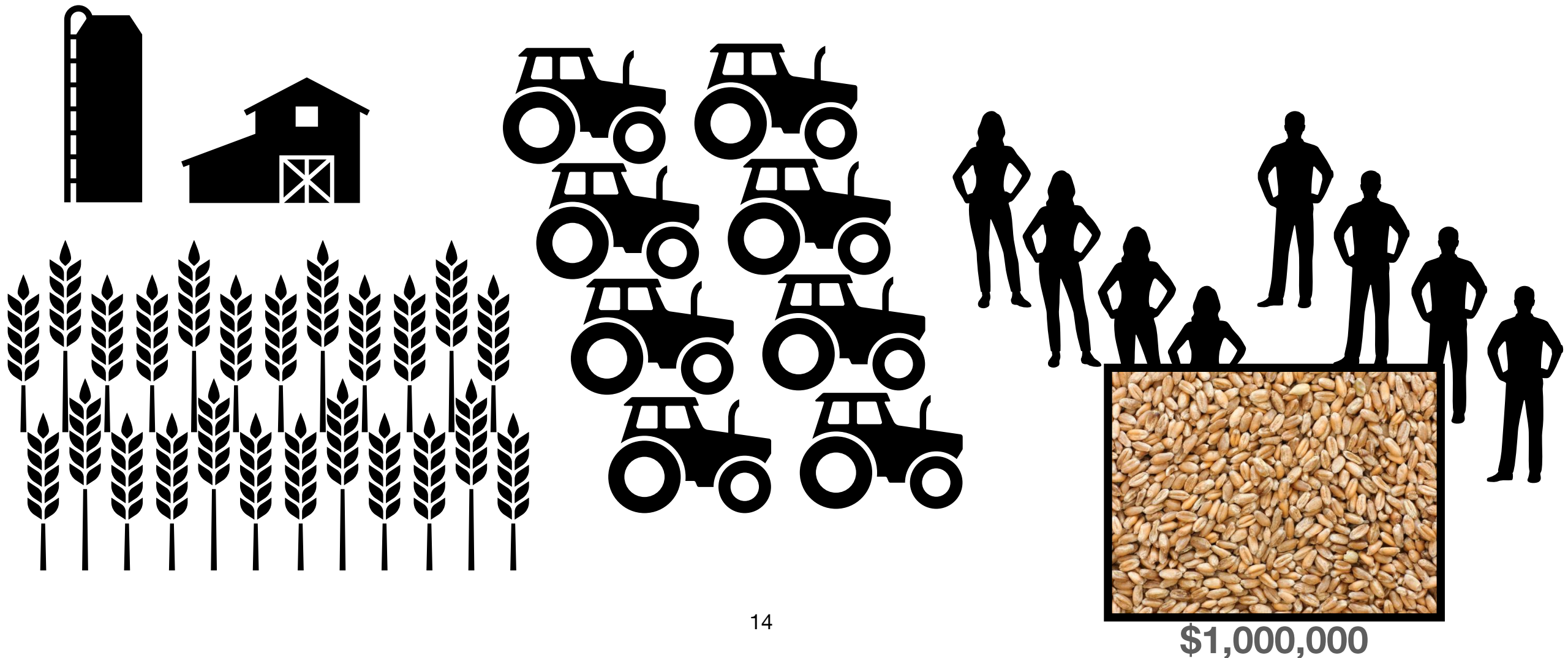
$$Y = \$1,000,000$$

$$Y = A K^{1/3} L^{2/3}$$

$$\$1,000,000 = 1,575 \cdot (4,000,000)^{1/3} (8)^{2/3}$$

$$\$1,000,000 = 1,575 \cdot 158.74 \cdot 4$$

The math is impossible to eyeball, but it checks out



# What the ...

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- What is  $A$ ?
- Exactly.
- It is often called
  - Total factor productivity (TFP)
  - [Multifactor productivity](#)



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- What is  $A$ ?
- Exactly.
- It is often called
  - Total factor productivity (TFP)
  - Multifactor productivity
- It isn't something we can actually measure
- Instead it's a **residual**
- Here, I knew what  $Y$ ,  $K$ , and  $L$  were, and I knew the production function. Then

$$A = \frac{Y}{K^{1/3} L^{2/3}}$$



# What drives TFP is a big bottom line in ECON 100B

$$K = \$4,000,000$$

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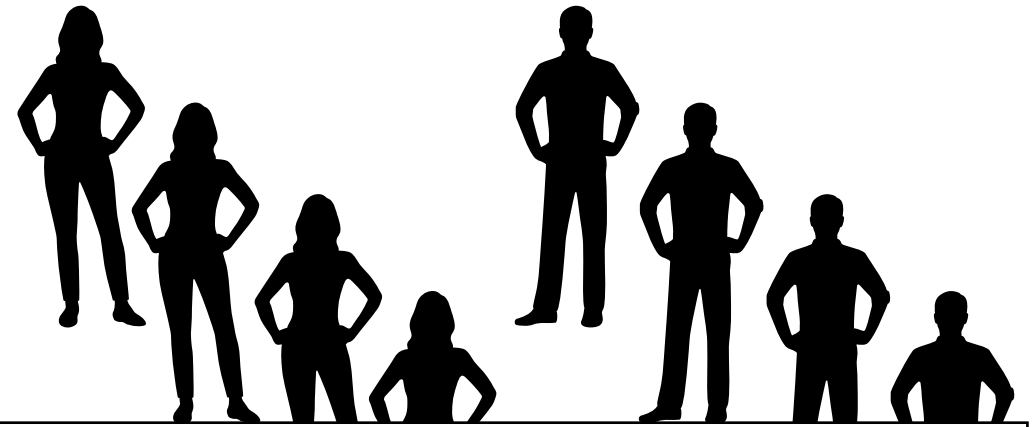
- Chapter 4 reveals that there must be really big differences across countries in the level of TFP, the variable  $A$
- Chapter 5 will show that capital  $K$  matters, but  $A$  matters more
- Chapter 6 will show us more about  $A$

But first, let's explore a little more



**Suppose we duplicated the farm**





- First farm:

$$Y_1 = A K_1^{1/3} L_1^{2/3}$$

$$\$1,000,000 = 1,575 \cdot (4,000,000)^{1/3} (8)^{2/3}$$

- Second farm:

$$Y_2 = A K_2^{1/3} L_2^{2/3}$$

$$\$1,000,000 = 1,575 \cdot (4,000,000)^{1/3} (8)^{2/3}$$

# Constant returns to scale

- For both farms combined, we have

$$Y_{total} = Y_1 + Y_2$$

$$Y_{total} = A K_1^{1/3} L_1^{2/3} + A K_2^{1/3} L_2^{2/3}$$

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- Therefore

$$Y_{total} = A \bar{K}^{1/3} \bar{L}^{2/3} + A \bar{K}^{1/3} \bar{L}^{2/3} = 2 A \bar{K}^{1/3} \bar{L}^{2/3}$$

$$Y_{total} = 2^{1/3} 2^{2/3} A \bar{K}^{1/3} \bar{L}^{2/3} = \boxed{A (2\bar{K})^{1/3} (2\bar{L})^{2/3}} = 2\bar{Y}$$

# Marginal products of capital

$$K = \$4,000,000$$

$$Y = A K^{1/3} L^{2/3}$$

$$L = 8$$

$$Y = \$1,000,000$$

$$\partial Y / \partial K = MPK = \frac{1}{3} A \left( \frac{L}{K} \right)^{2/3}$$

$$= \frac{1}{3} 1,575 \left( \frac{8}{\$4,000,000} \right)^{2/3}$$

$$= 0.0833$$

This is like an *interest rate*

Here, it is the gross rate of  
return on risky capital  
investment



# Marginal products of capital and labor

$$K = \$4,000,000$$

$$L = 8$$

$$Y = \$1,000,000$$

$$Y = A K^{1/3} L^{2/3}$$

$$\partial Y / \partial K = MPK = \frac{1}{3} A \left( \frac{L}{K} \right)^{2/3}$$

$$= \frac{1}{3} 1,575 \left( \frac{8}{\$4,000,000} \right)^{2/3}$$

$$= 0.0833$$

$$\partial Y / \partial L = MPL = \frac{2}{3} A \left( \frac{K}{L} \right)^{1/3}$$

$$= \frac{2}{3} 1,575 \left( \frac{\$4,000,000}{8} \right)^{1/3}$$

$$= \$83,333$$

This is an *annual salary*

Across 8 workers, it sums to \$66,666.67 or two thirds of GDP



## Part 2 of Chapter 4: Explaining GDP across countries

- How should we do this? It's challenging
- Suppose we measured  $Y$ ,  $K$ ,  $L$ , and inferred  $A$  for each country  
... congratulations, you described reality. Now what?
- Well, maybe we could examine  $A$  across countries and see how crazy it looks. As a residual, it is true by definition. But is it crazy?
- What does “crazy” mean here? Something that is **orders of magnitude different** between otherwise similar countries on earth

## Side note: Constant returns to scale provides us a nifty trick

- It turns out that  $\frac{Y}{L} \equiv y = \frac{A K^{1/3} L^{2/3}}{L} = A k^{1/3}$  for  $k \equiv \frac{K}{L}$

in words:

Income per worker  $y$  is a concave function of capital per worker  $k$

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in words:

Income per worker  $y$  is a concave function of capital per worker  $k$

- Now use your **growth rate rules**

$$g[y] = g[A] + \frac{1}{3}g[k]$$

- If we see a country with 6% less capital per worker, we would expect that its income per worker is 2% less ... plus any effect of total factor productivity (TFP)

# Without TFP differences, the model doesn't do well

TABLE 4.3

## The Model's Prediction for Per Capita GDP (U.S. = 1)

| Country       | Observed capital per person, $\bar{k}$ | Predicted per capita GDP<br>$y = \bar{k}^{1/3}$ | Observed per capita GDP |
|---------------|----------------------------------------|-------------------------------------------------|-------------------------|
| United States |                                        |                                                 |                         |
| Switzerland   |                                        |                                                 |                         |
| Japan         |                                        |                                                 |                         |
| Italy         |                                        |                                                 |                         |
| Spain         |                                        |                                                 |                         |
| U.K.          |                                        |                                                 |                         |
| Brazil        |                                        |                                                 |                         |
| China         |                                        |                                                 |                         |
| South Africa  |                                        |                                                 |                         |
| India         |                                        |                                                 |                         |
| Burundi       |                                        |                                                 |                         |

Here,  $\bar{k}$  for each country is divided by the U.S. level

So if the number you see is 0.914, that means capital per worker is 91.4% of the U.S. level

And a model with fixed  $A = 1$  predicts  $y = 0.914^{1/3} = 0.971$  or 97.1% of the U.S. level

Predicted per capita GDP is computed as  $\bar{k}^{1/3}$ , that is, assuming no differences in productivity across countries. Data correspond to the year 2019 and are divided by the values for the United States.

Source: Penn World Table, Version 10.0.

# Without TFP differences, the model doesn't do well

TABLE 4.3

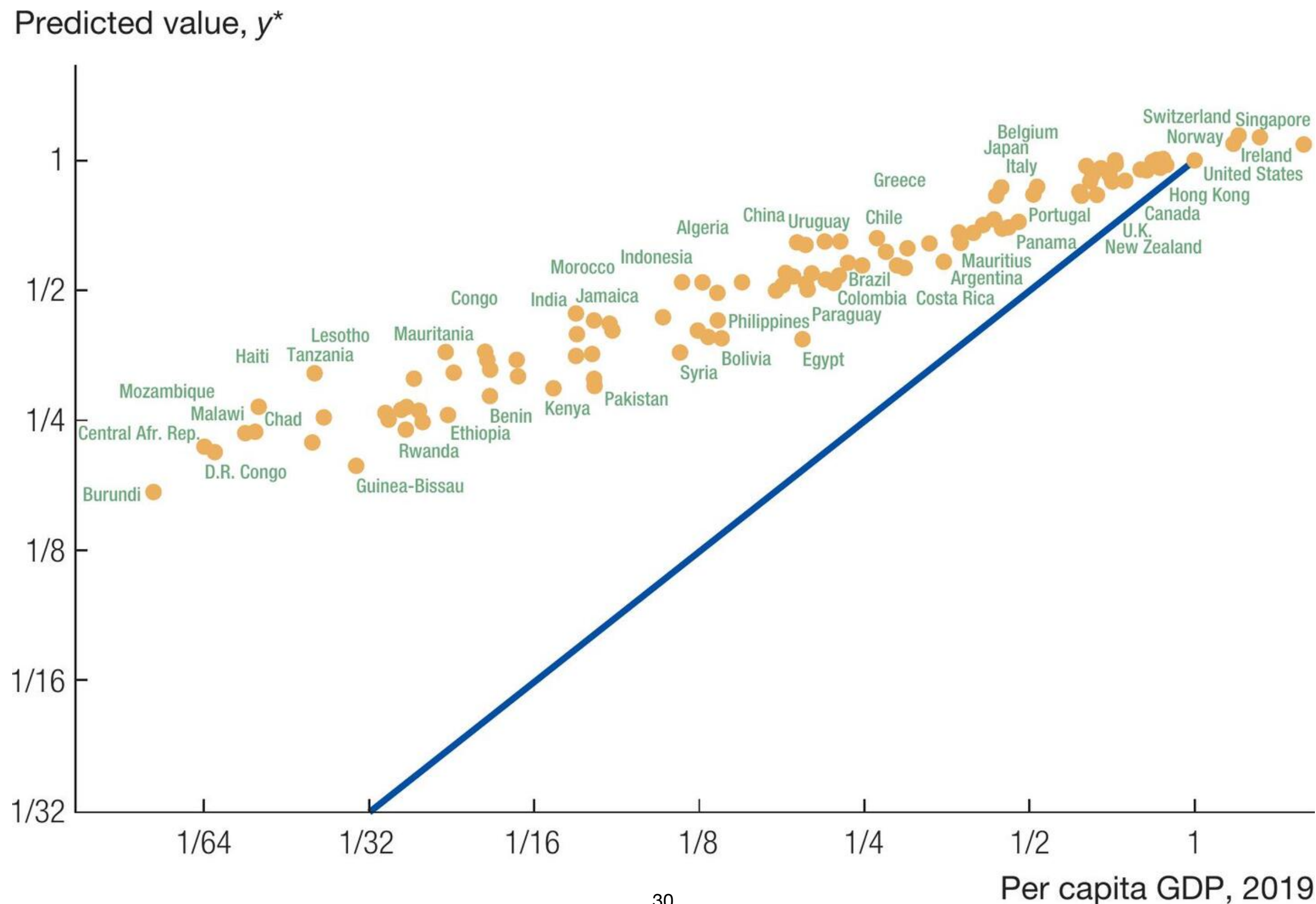
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|---------------|----------------------------------------|-------------------------------------------------|-------------------------|
| United States | 1.000                                  | 1.000                                           | 1.000                   |
| Switzerland   | 1.488                                  | 1.142                                           | 1.203                   |
| Japan         | 0.914                                  | 0.971                                           | 0.634                   |
| Italy         | 0.792                                  | 0.925                                           | 0.651                   |
| Spain         | 0.716                                  | 0.895                                           | 0.645                   |
| U.K.          | 0.714                                  | 0.894                                           | 0.707                   |
| Brazil        | 0.195                                  | 0.580                                           | 0.233                   |
| China         | 0.275                                  | 0.650                                           | 0.226                   |
| South Africa  | 0.165                                  | 0.548                                           | 0.200                   |
| India         | 0.082                                  | 0.434                                           | 0.107                   |
| Burundi       | 0.005                                  | 0.172                                           | 0.013                   |

Predicted per capita GDP is computed as  $\bar{k}^{1/3}$ , that is, assuming no differences in productivity across countries. Data correspond to the year 2019 and are divided by the values for the United States.

Source: Penn World Table, Version 10.0.

# Without TFP differences, most predictions undershoot reality. But Switzerland, Ireland, Singapore are overperformers





# When we compute TFP as the residual, we see big differences across countries

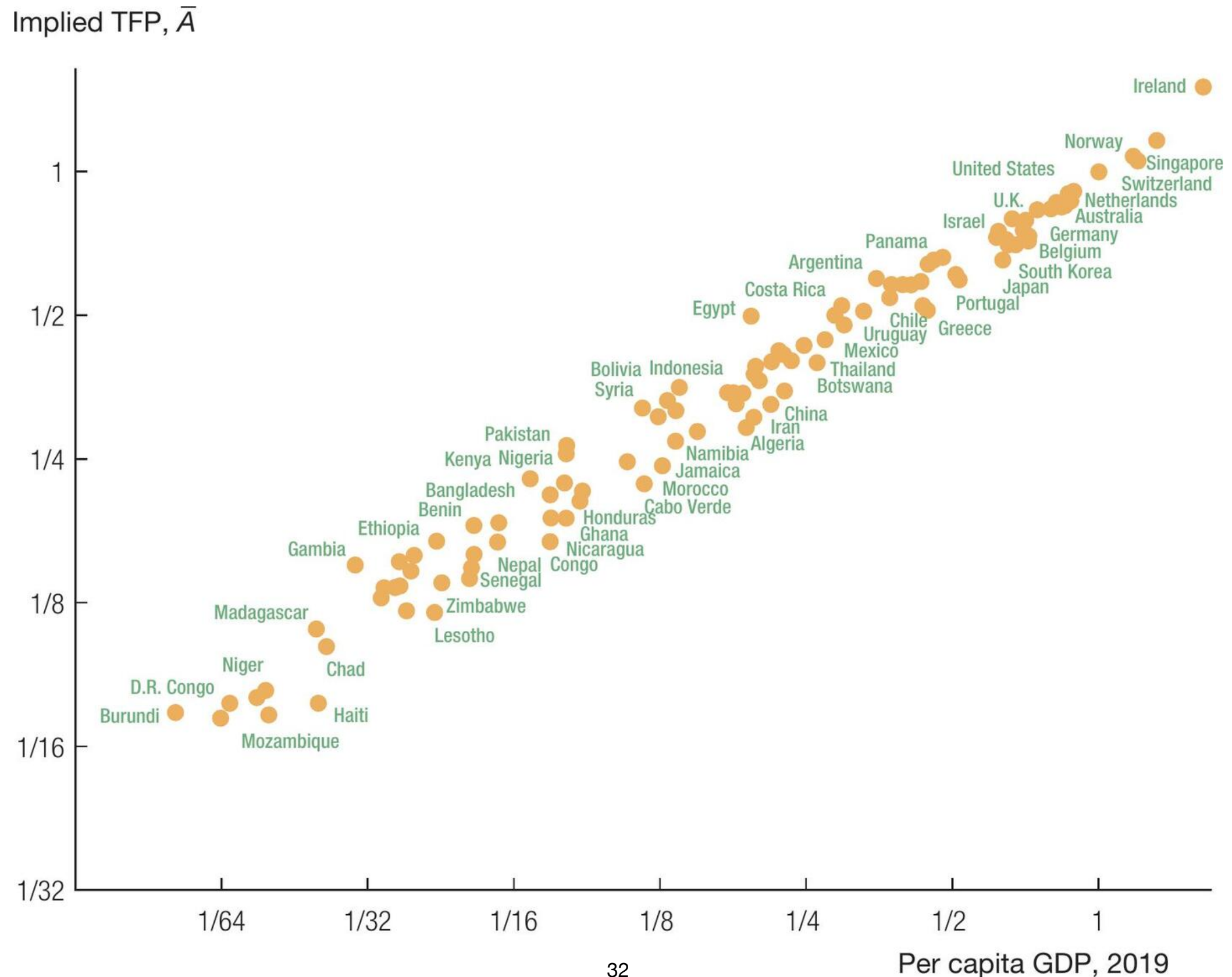
TABLE 4.4

## Measuring TFP So the Model Fits Exactly

| Country       | Per capita<br>GDP ( $y$ ) | $\bar{k}^{1/3}$ | Implied<br>TFP ( $\bar{A}$ ) |
|---------------|---------------------------|-----------------|------------------------------|
| United States | 1.000                     | 1.000           | 1.000                        |
| Switzerland   | 1.203                     | 1.142           | 1.054                        |
| U.K.          | 0.707                     | 0.894           | 0.792                        |
| Japan         | 0.634                     | 0.971           | 0.654                        |
| Italy         | 0.651                     | 0.925           | 0.703                        |
| Spain         | 0.645                     | 0.895           | 0.721                        |
| Brazil        | 0.233                     | 0.580           | 0.402                        |
| South Africa  | 0.200                     | 0.548           | 0.365                        |
| China         | 0.226                     | 0.650           | 0.347                        |
| India         | 0.107                     | 0.434           | 0.247                        |
| Burundi       | 0.013                     | 0.172           | 0.074                        |

Calculations are based on the equation  $y = \bar{A}\bar{k}^{1/3}$ . Implied productivity  $\bar{A}$  is calculated from data on  $y$  and  $\bar{k}$  for the year 2019, so that this equation holds exactly as  $\bar{A} = y/\bar{k}^{1/3}$ .

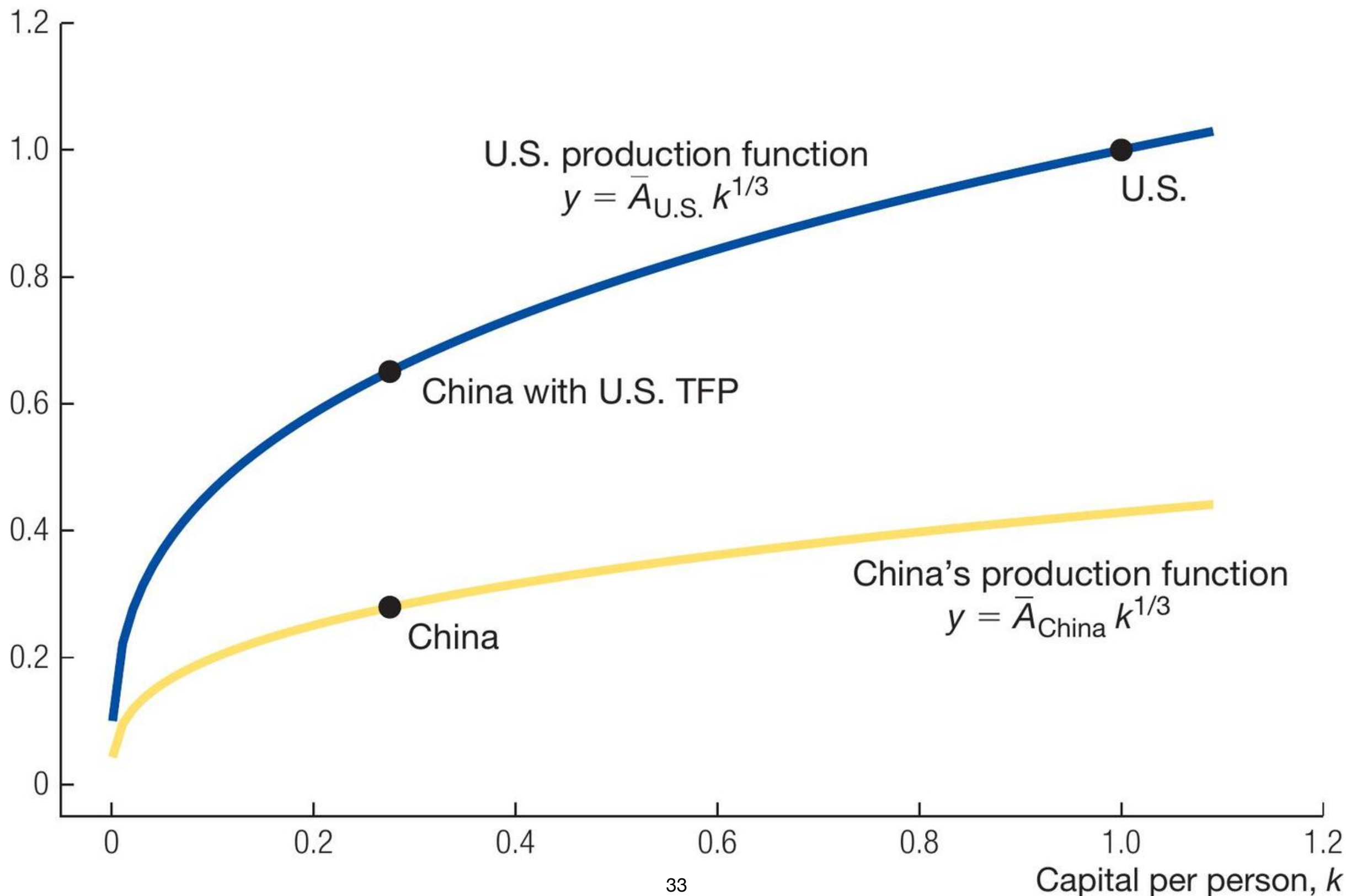
# TFP across countries is a “luxury good,” rising strongly with income per person





# TFP matters a lot. It shifts the production function upward, raising output per person as a function of capital per person

Output per person,  $y$



# Teaser for Chapter 6 and Nobel Prize

Good income measures for North Korea are not readily available. However, this image of nighttime light output vividly conveys the relative income differences between North and South Korea. The obvious explanation is institutional differences. But which institutions, exactly?





Like a Barnaby?

Best [mondegreen](#) ever?

